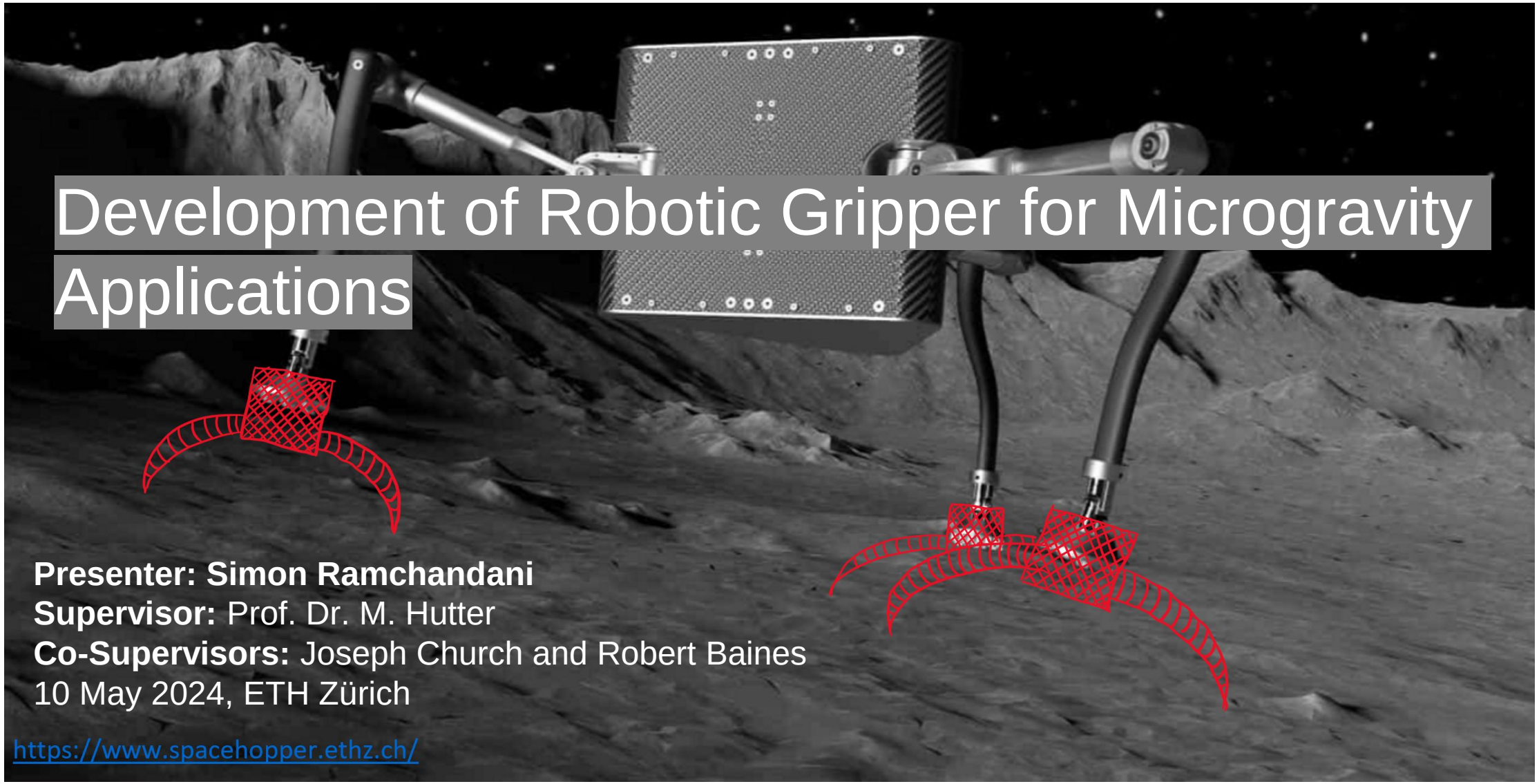
ETH zürich

RSL Robotic Systems Lab

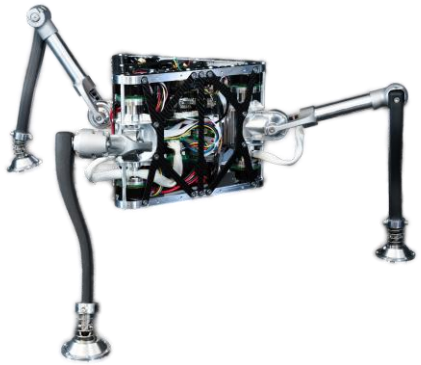


Development of Robotic Gripper for Microgravity Applications

Presenter: Simon Ramchandani
Supervisor: Prof. Dr. M. Hutter
Co-Supervisors: Joseph Church and Robert Baines
10 May 2024, ETH Zürich

https://www.spacehopper.ethz.ch/

Motivation for the whole projects



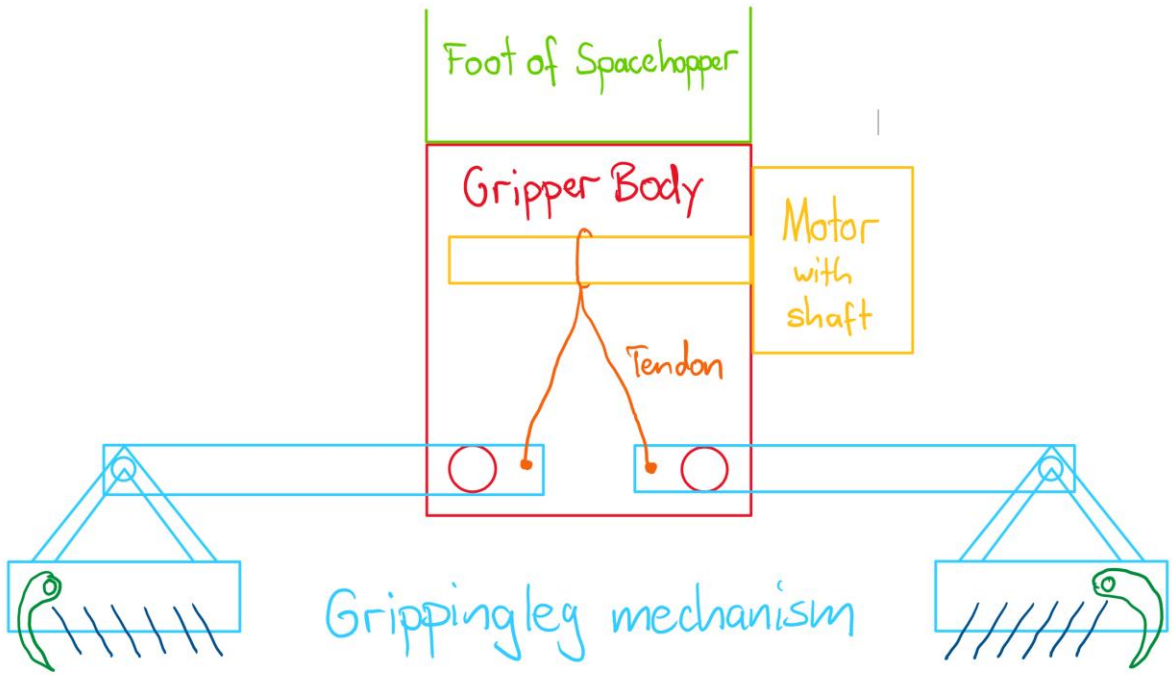
- Exploring surface of asteroids and other microgravitational environments

- Design a robotic gripper for robots like Spacehopper

- Considerations:

  - grasp and hold on to rocks
  - should work in zero gravity
  - designed for space

Overview and division of workload



Fiona
- Design of gripping mechanism

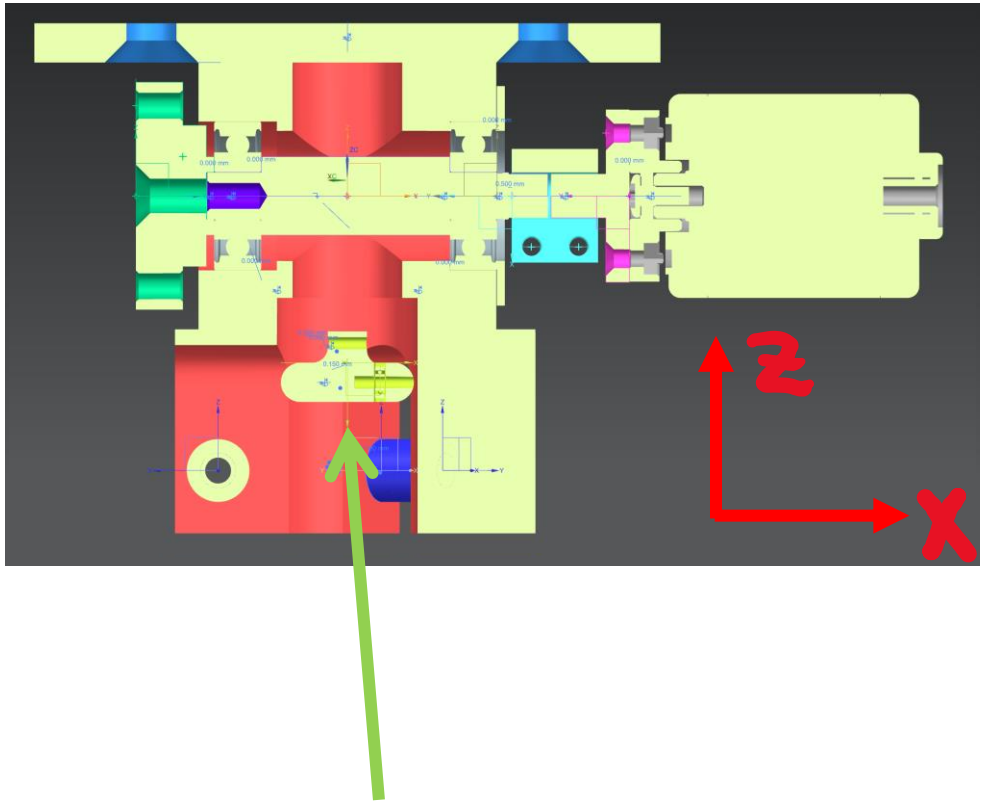
Simon
- Design of gripper body
- Actuation of gripper
- Setup and conducting tests:

  -selecting rock to gripp

  - determining the capable adhesion force in tangential and normal direction

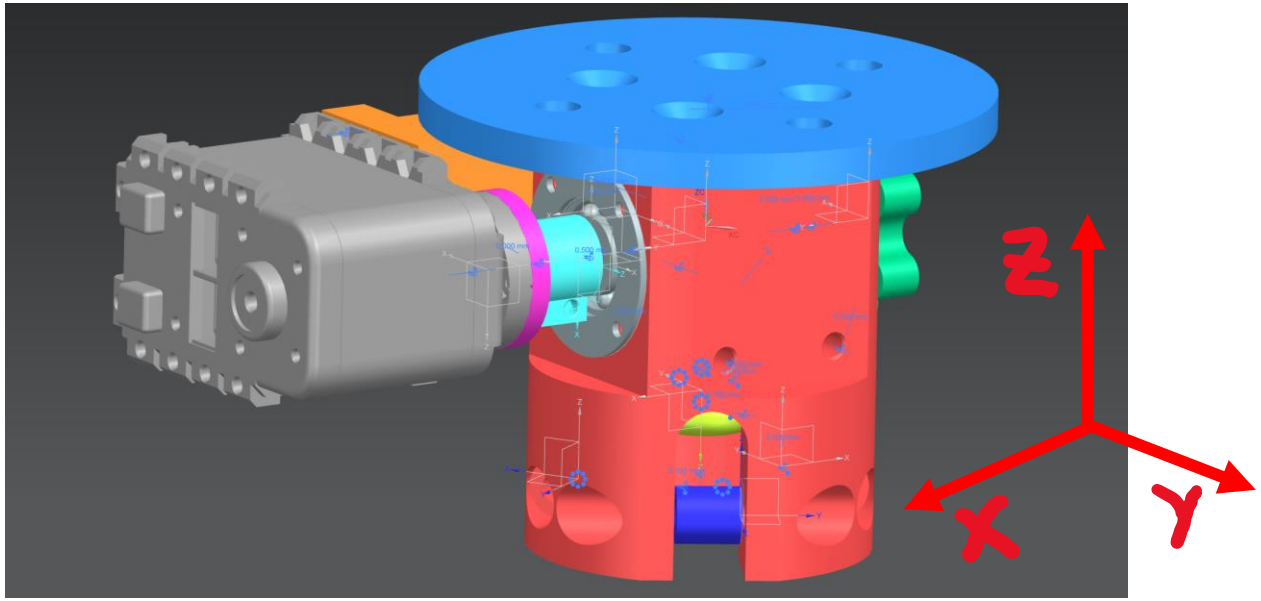
  - test different hooking mechanisms

Gripper body Design

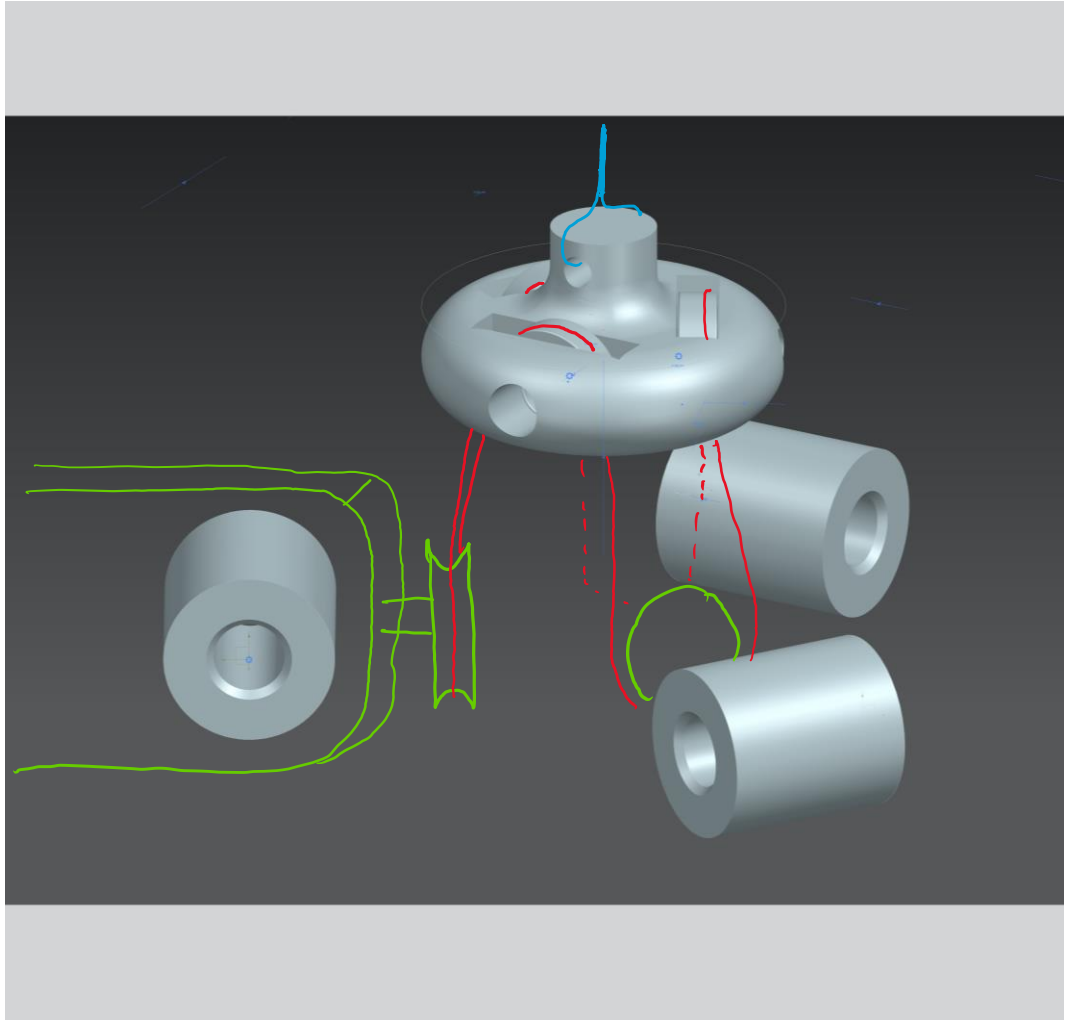


- Differential mechanism

- Motor drives the shaft and pulls up tendon
- Gripper can be closed by hand (for test purpose)
- Motor shaft pulls up tendon that is attached to a differential mechanism

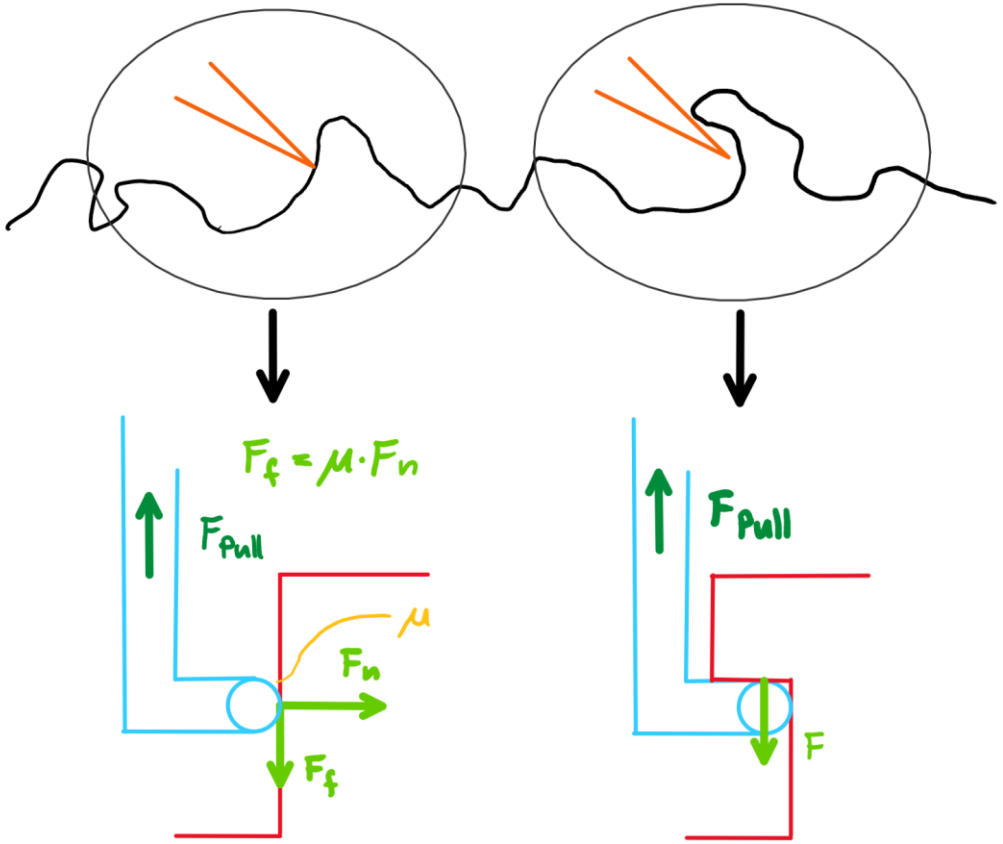


Gripper body Design – Differential mechanism



- Blue tendon gets pulled up by the motor
- If one of the gripper legs hooks onto the rock, the other ones will keep moving up due to the differential mechanism
- Equal foce on each gripping leg
- Red tendon is a closed loop
- Redesign form to prevent tilting and rotating

Testing – Two main hooking mechanism



- 1. Mechanism based on friction (left)

  The bigger the generated **force in tangential direction**, the bigger the the capable pulling force

- 2. Mechanism based on geometry (right)

  For this the geometry and angle of the spine is the main contributing factor

  (a force in tangential direction is necessary to keep the spine in his place)

Test setup in the literature

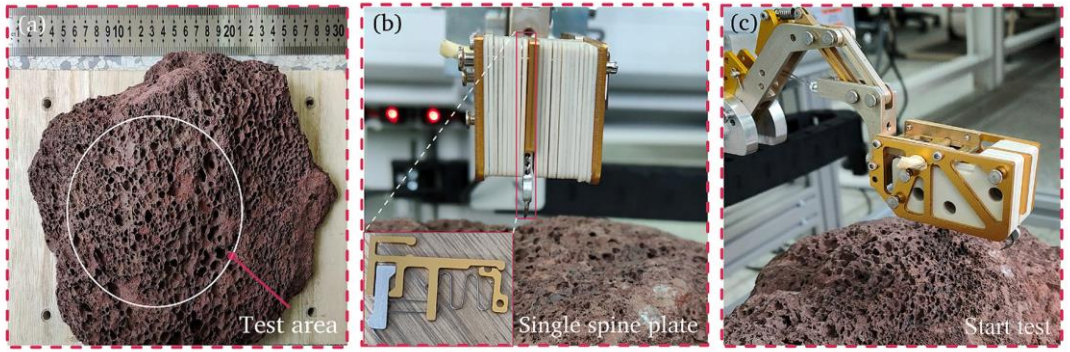


Fig. 8. Adhesion test process. (a) Rock used in the test; (b) Preparation before testing; (c) Testing in progress.

Fig. 8: Q. Cheng, W. Zhang, W. Zhou, J. Li, Z. Li, T. Yu, B. Wang, and S. Wang, “Design and analysis of a bionic adhesion coring sampler for space unstructured surface,” Acta Astronautica, vol. 215, pp. 79–101, Feb. 2024

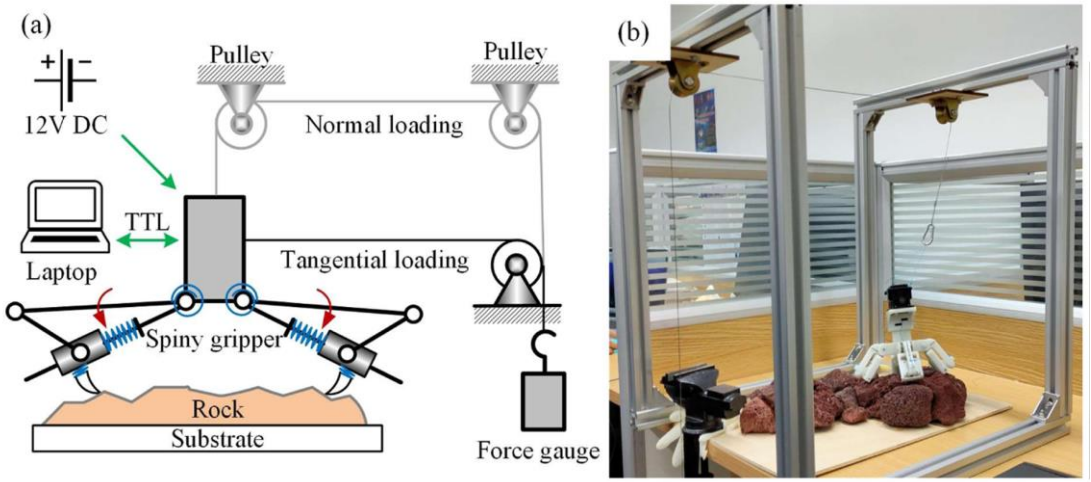


Fig.: P. Zi, K. Xu, Y. Tian, and X. Ding, “A mechanical adhesive gripper inspired by beetle claw for a rock climbing robot,” Mechanism and Machine Theory, vol. 181, p. 105168, Mar. 2023

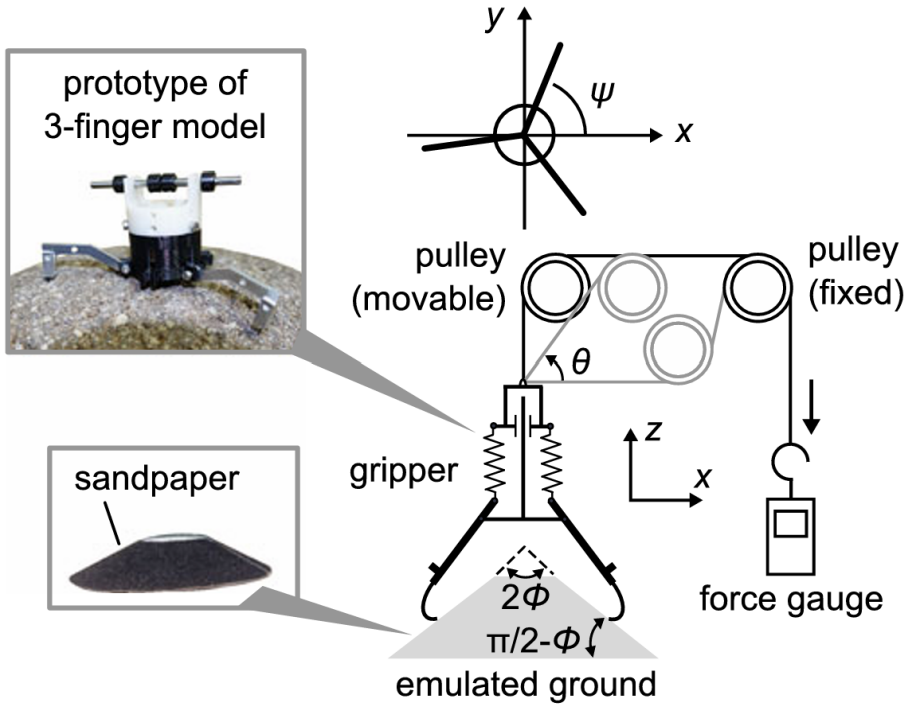
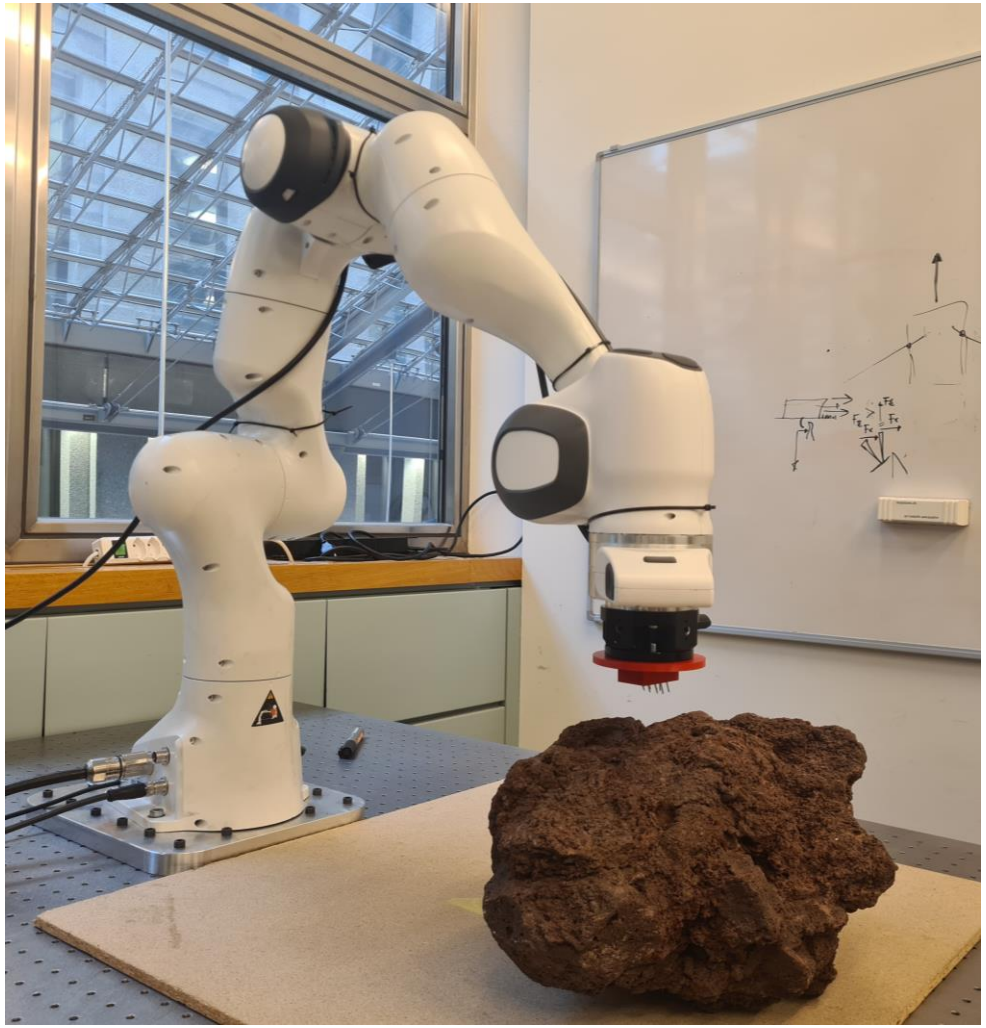


Fig.:K. Nagaoka, H. Minote, K. Maruya, Y. Shirai, K. Yoshida, T. Hakamada, H. Sawada, and T. Kubota, “Passive Spine Gripper for Free-Climbing Robot in Extreme Terrain,” IEEE Robotics and Automation Letters, vol. 3, pp. 1765–1770, July 2018. Conference Name: IEEE Robotics and Automation Letters

Test setup



- Gripper or subpart of gripper attached to Franka arm with force-torque-sensor

  versatile in the movement of the arm

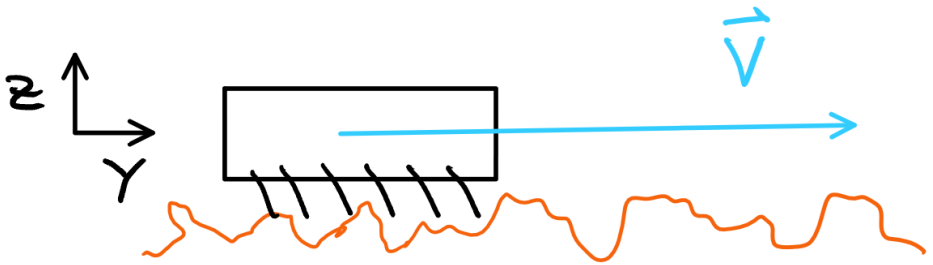
  measure forces and torques in all directions

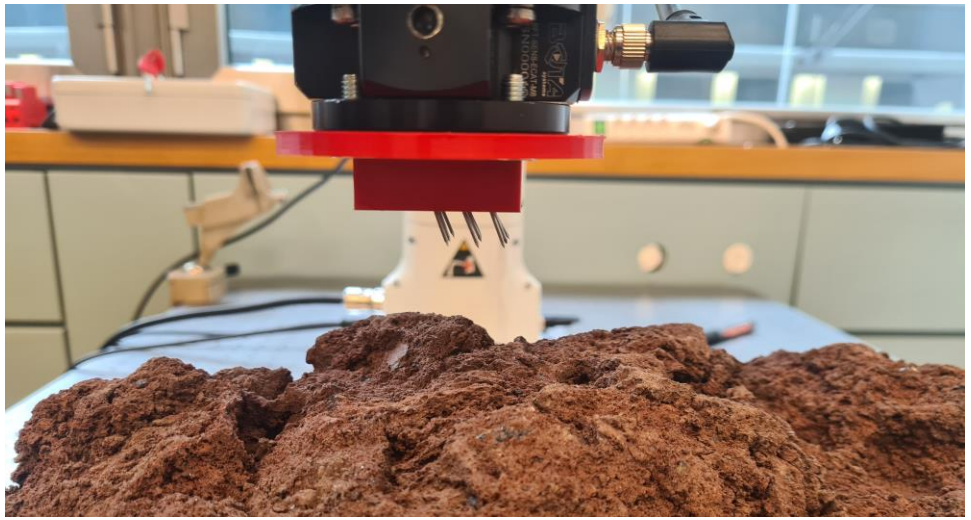
- Decided to test on **porous volcanic rock**

  exists widely on mars and other numerous planets

  unregular surface to capture as many surface structures as possible

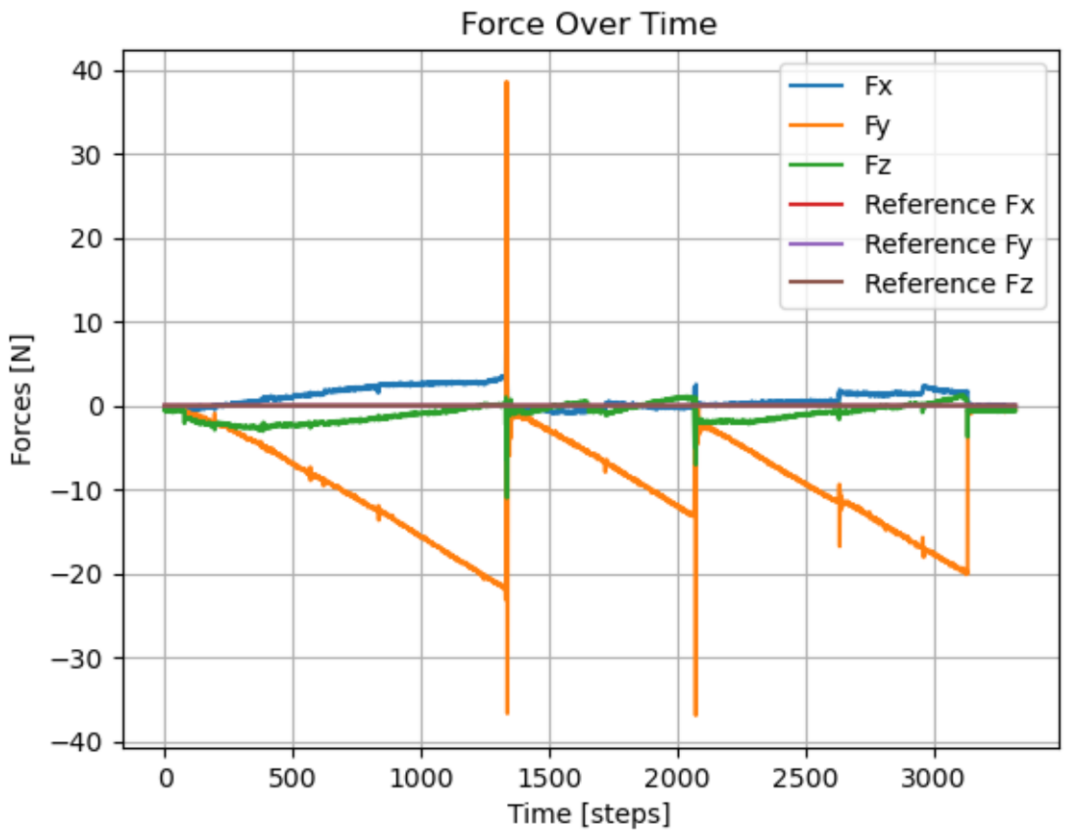
Testing tangential force (y-force)

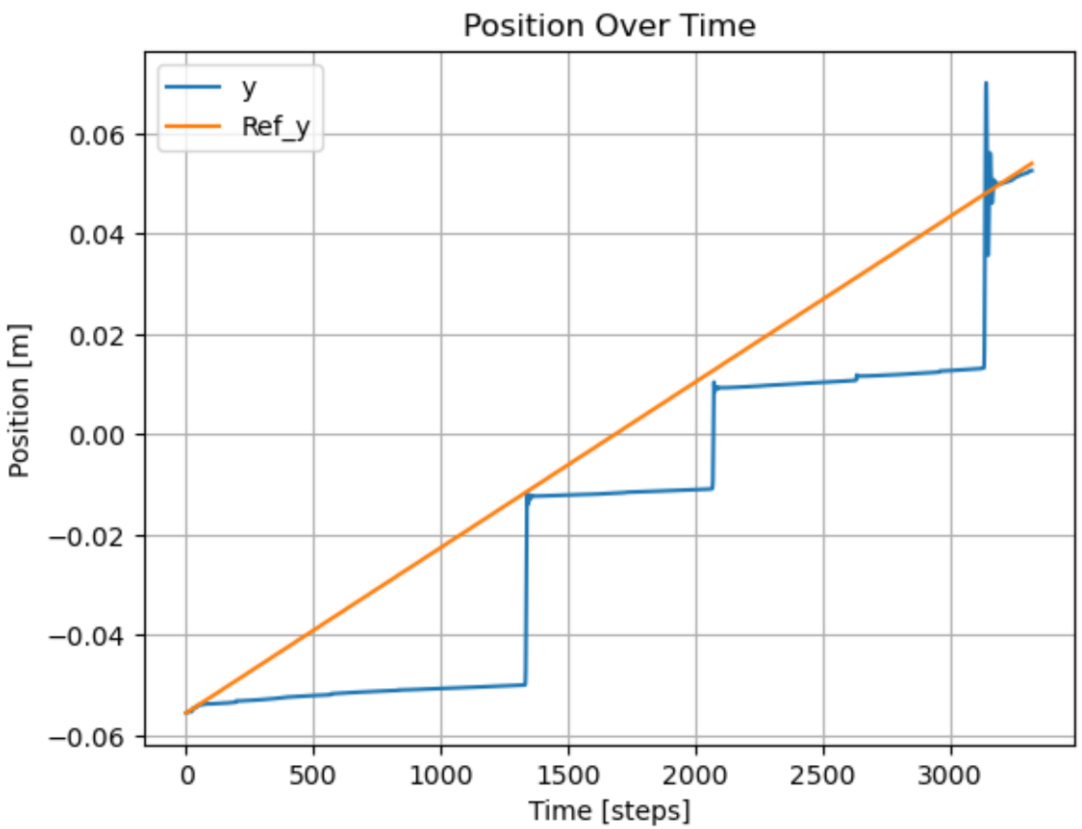




- Track velocity in y-direction with Franka-Arm
- Measure the forces in x, y and z direction
- Primarily interested in the capable y-force bevor detaching
- Test different spine angles

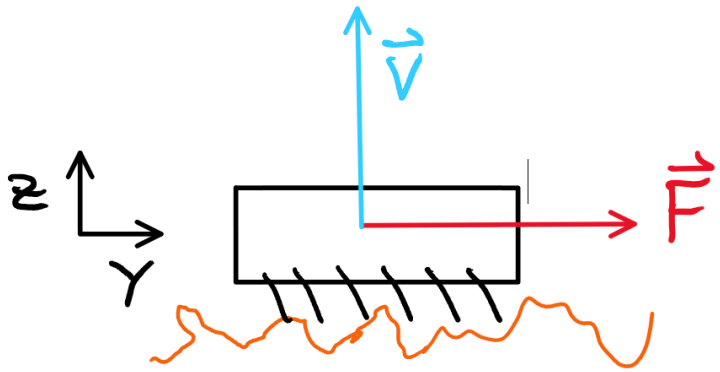
Testing tangential force (y-force)



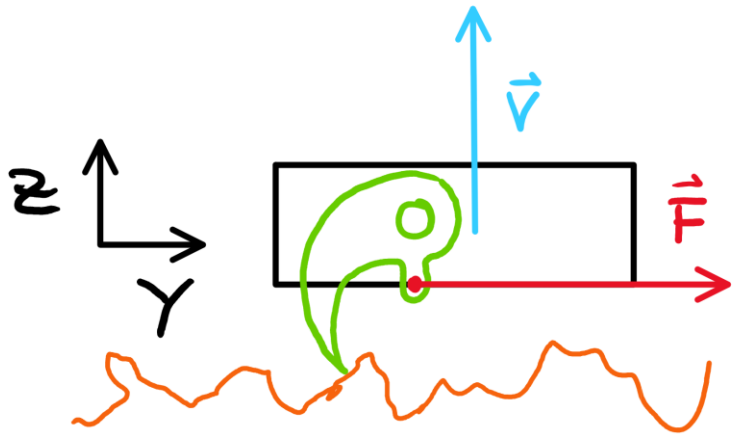


- Approximately 10 to 20 Newtons before detaching

Future Testing – pulling force and hook implementation



- Apply force in y direction and velocity in z direction
- Measure the z force (pulling force)



- Test hook mechanism with different configurations
- Also measure the z force

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